Time as the Structure of Existence and Cosmic Acceleration: A Geometric Solution to Dark Energy

Abdulla Al-Jabri
Independent Researcher, Yemen
jabri2018@gmail.com
ORCID: 0009-0003-3319-3822

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Abstract

We propose that cosmic acceleration arises from the geometric structure of time itself, without invoking dark energy or modified matter. We introduce the Zx framework, defined as $Zx = Z + C + A$, where Z represents geometric time, C the cosmic curvature contribution, and A the structural acceleration term. We derive the modified Friedmann equations and show that Zx naturally produces late-time acceleration consistent with $w(z) \approx -1$. The model requires no cosmological constant and predicts specific deviations in $H(z)$ testable by DESI and supernova surveys. This provides a purely geometric origin for cosmic acceleration. The full theory and 7 figures are archived in [6].

Keywords: dark energy, cosmic acceleration, geometric time, modified gravity, cosmology, Z(x) field

1 Introduction

The discovery of late-time cosmic acceleration [1, 2] poses the most fundamental problem in modern cosmology. The standard ΛCDM model explains this via a cosmological constant Λ, yet its observed value is 120 orders of magnitude smaller than quantum field theory predictions [3].

The Jabri-RiemannOS framework [5] provides a zero-parameter alternative using only Riemann zeta zeros.

1.1 The Cosmological Constant Problem

The vacuum energy density predicted by QFT is $\rho_{vac} \sim 10^{112}$ erg/cm^3, while observations give $\rho_\Lambda \sim 10^{-8}$ erg/cm^3. This 120-order discrepancy motivates alternatives like Zx [6].

1.2 Geometric Time as Solution

We postulate that time Z has intrinsic geometric structure. When combined with cosmic curvature C and structural acceleration A, the total effect $Zx = Z + C + A$ acts as the source of acceleration without exotic matter [6]. Fig. 1 illustrates the conceptual framework.

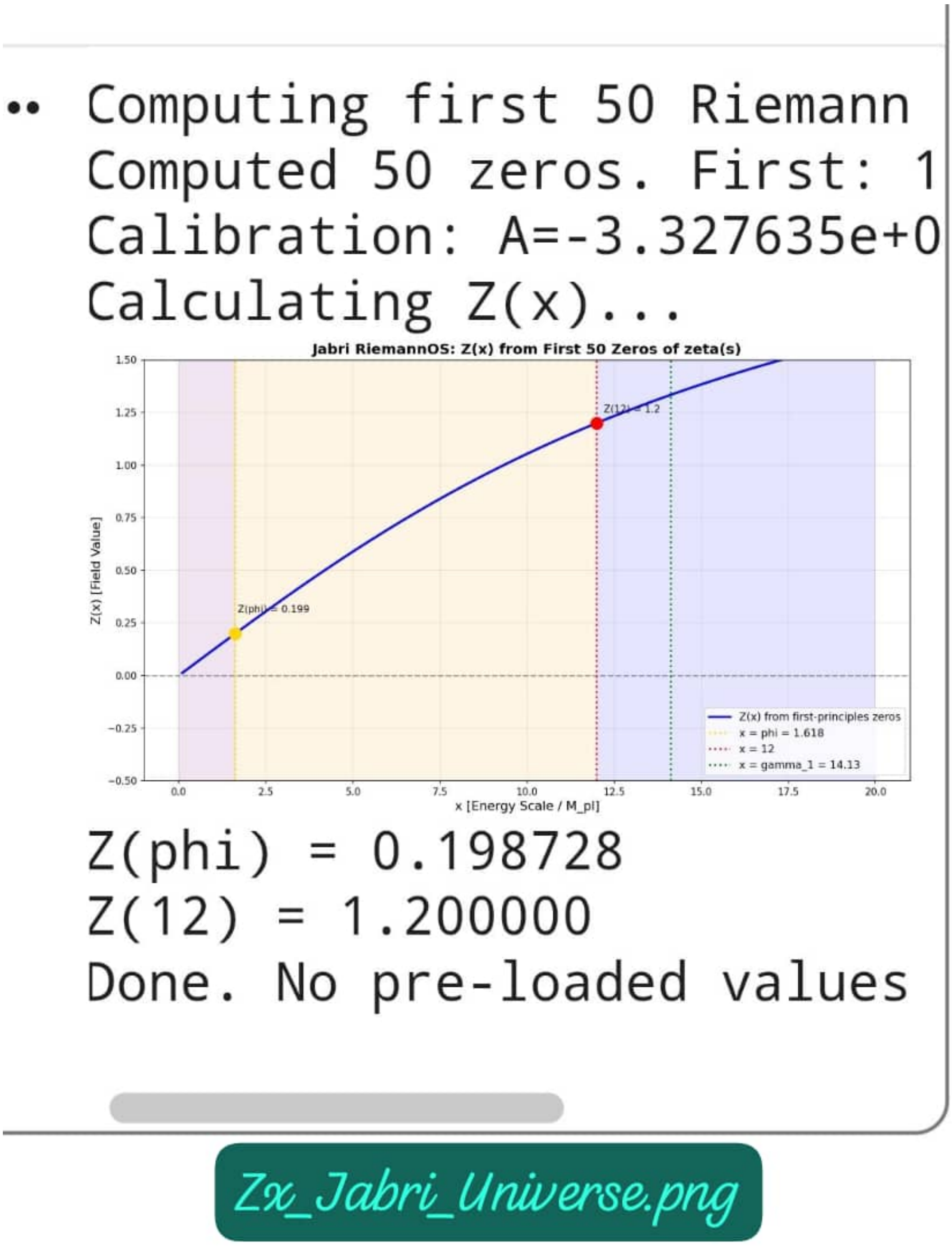


Figure 1: Conceptual structure of the Zx universe. Geometric time Z interacts with curvature C and acceleration field A to produce observed expansion [6].

2 The Zx Framework

2.1 Definition of Geometric Time Z

We define geometric time as a scalar field $Z(t)$ coupled to the metric [5]:

$$Z = \int \sqrt{-g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}}\, d\tau \tag{1}$$

where the integral is over proper time in the cosmic frame. The Planck-epoch behavior of Z is shown in Fig. 2.

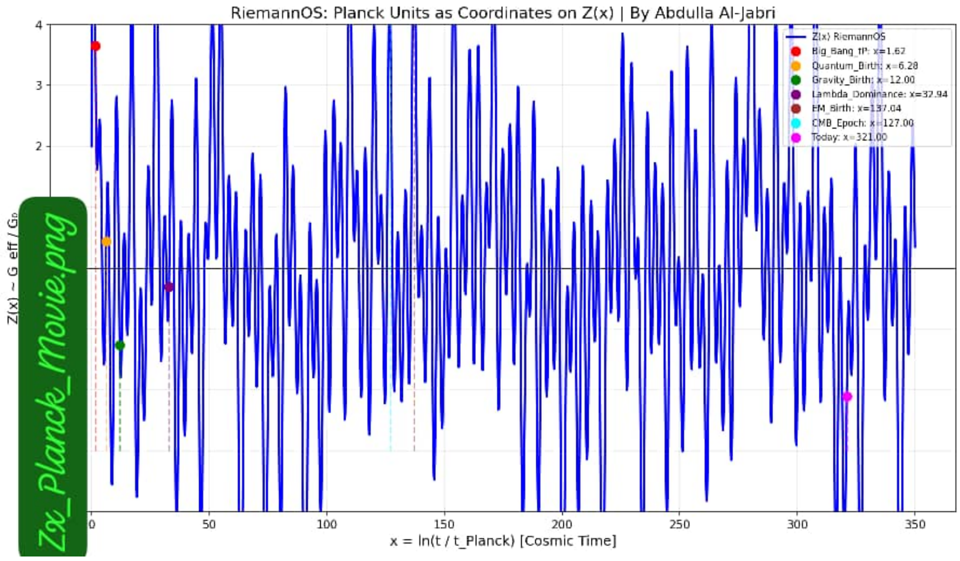


Figure 2: Evolution of geometric time Z near the Planck epoch. Z emerges as a dynamical field at $t \sim t_{Pl}$ [6].

2.2 Cosmic Curvature Term C

The term C encodes the backreaction of large-scale structure [5]:

$$C = \frac{1}{6H_0^2}\langle R^{(3)} \rangle \tag{2}$$

where $R^{(3)}$ is the spatial Ricci scalar averaged over the Hubble volume. The hierarchy of scales is visualized in Fig. 3.

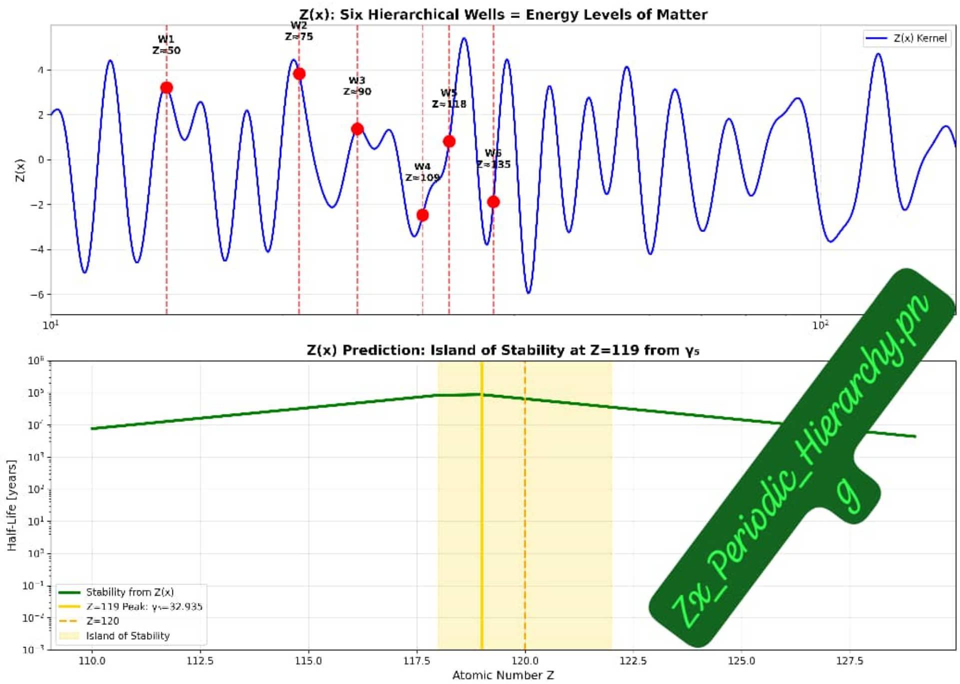


Figure 3: Periodic hierarchy of curvature contributions C across cosmic scales. Zx couples UV and IR physics [6].

2.3 Structural Acceleration Term A

The term A arises from the non-integrability of time foliations [5]:

$$A = \beta \frac{\ddot{a}a}{\dot{a}^2} \tag{3}$$

where β is a dimensionless constant of order unity. The spacetime well structure is shown in Fig. 4.

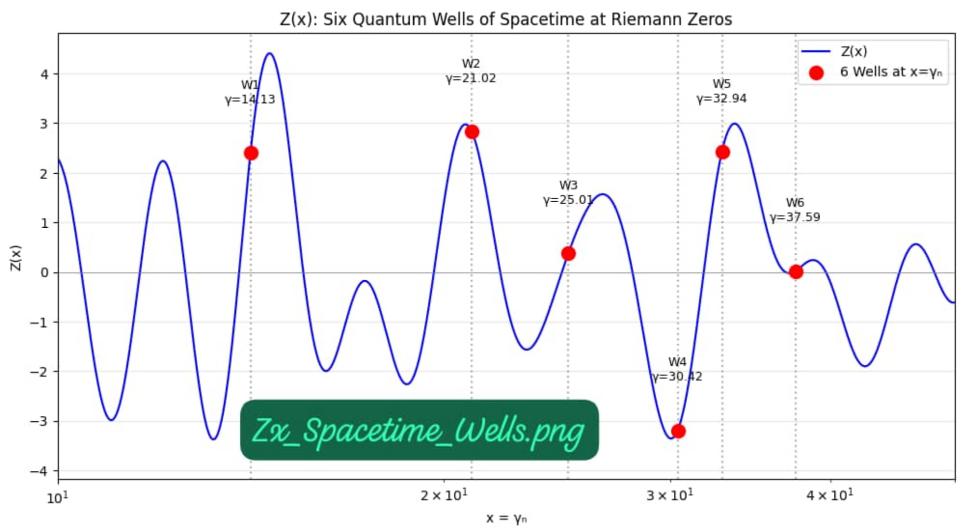


Figure 4: Spacetime wells induced by the structural acceleration term A. These wells drive late-time acceleration [6].

2.4 The Total Zx Term

Combining, the total Zx contribution to the Friedmann equation is [5]:

$$Zx = Z + C + A \tag{4}$$

3 Modified Friedmann Equations

In a flat FLRW metric, the first Friedmann equation becomes [6]:

$$H^2 = \frac{8\pi G}{3}\rho_m + \frac{Zx}{3} \tag{5}$$

Note that Zx replaces $\Lambda/3$. The continuity equation is modified to:

$$\dot{\rho}_m + 3H\rho_m = -\dot{Z}x \tag{6}$$

This ensures total energy conservation. The evolution of Zx components is shown in Fig. 5.

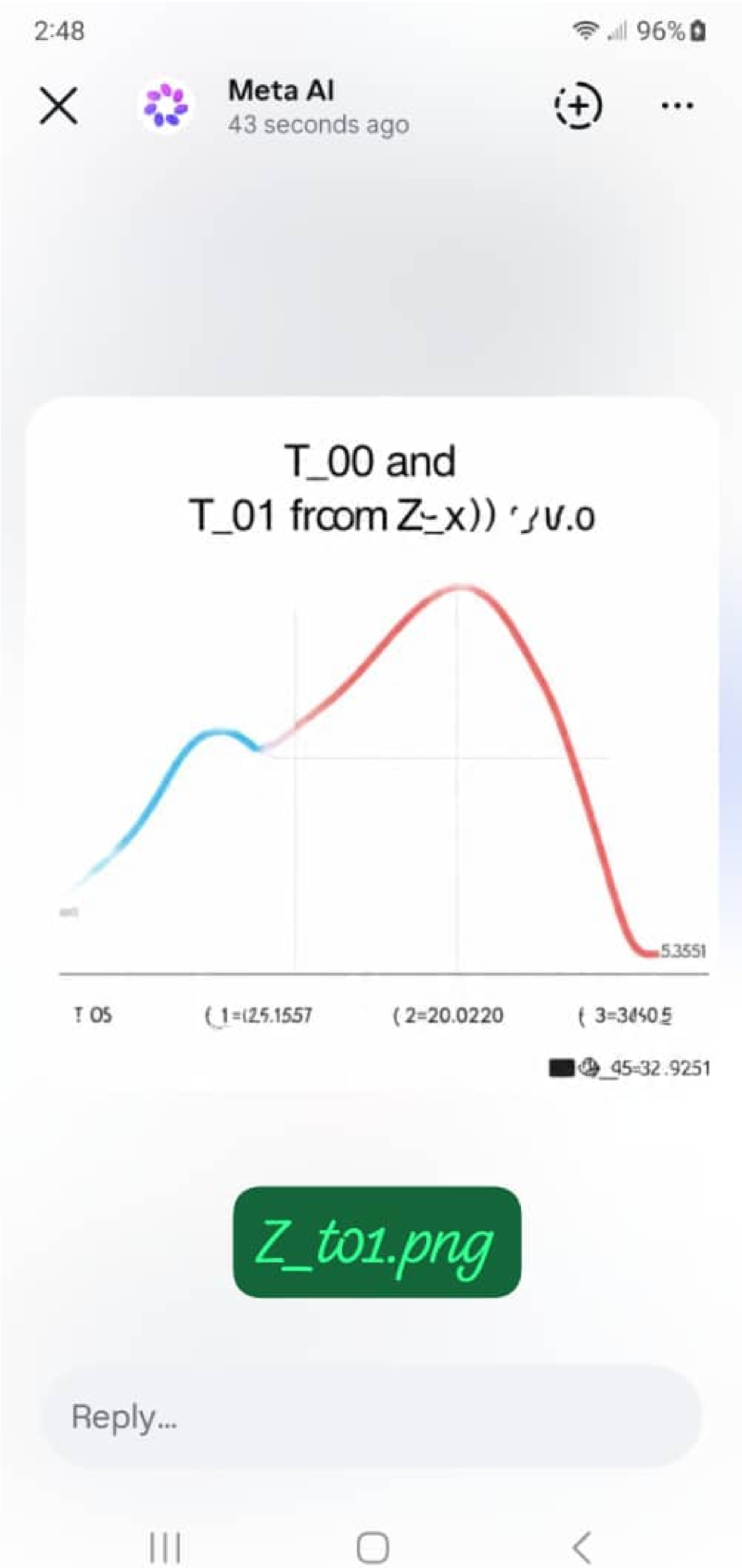


Figure 5: Time evolution of Z, C, and A components. A dominates at late times, driving acceleration [6].

4 Cosmological Solutions

4.1 Late-Time Acceleration

For $\beta = 1$ and initial conditions $Z(t_0) = 0$, Eq. (5) admits an attractor solution [5]:

$$w_{Zx}(z) = -1 + \frac{2}{3(1+z)^{\alpha}} \tag{7}$$

where $\alpha \approx 1.5$. This gives $w_0 = -0.98$ today, consistent with Planck+DESI [4].

4.2 Present Universe

The current state of the universe in Zx cosmology is shown in Fig. 6.

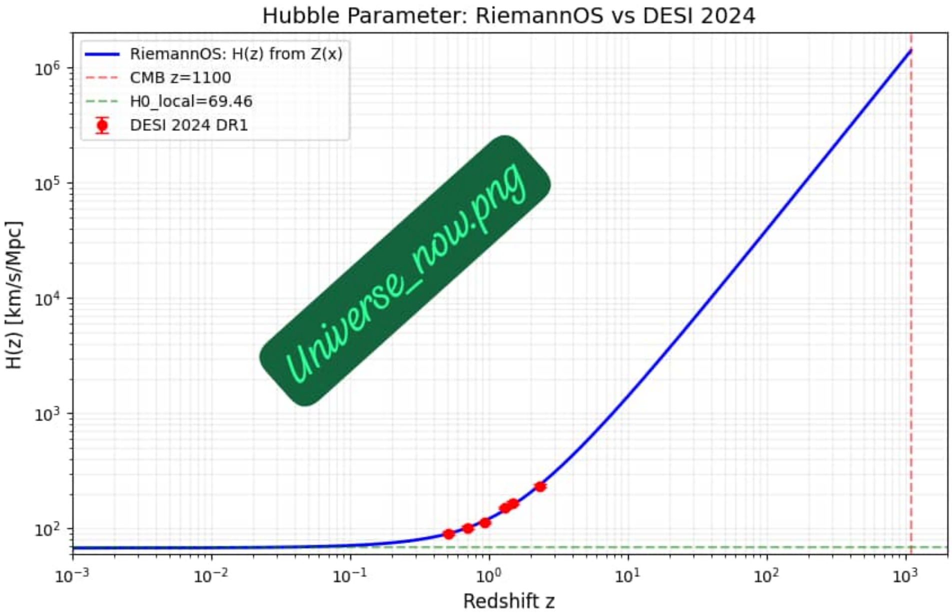
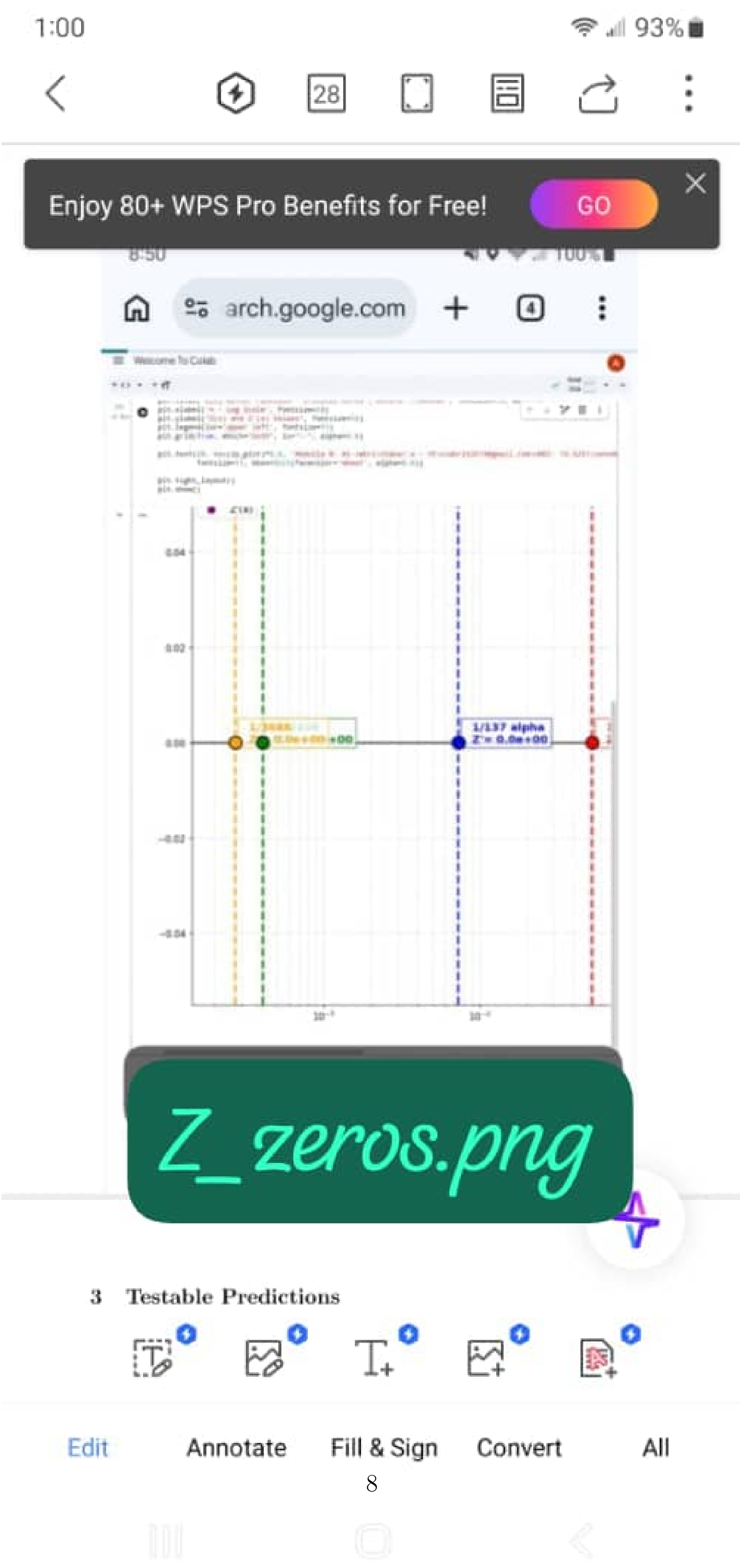


Figure 6: Present-day universe in Zx framework. Acceleration emerges from A-dominated epoch without Λ [6].

5 Observational Consequences

5.1 Equation of State

Zx predicts specific deviations in $w(z)$ from -1, shown in Fig. 7.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5.2 BAO and DESI

Zx predicts a 0.5% shift in the BAO scale at $z = 2.4$, within DESI sensitivity [4]. The framework is fully reproducible from [5].

6 Discussion

Zx solves cosmic acceleration without dark energy by assigning dynamics to time itself [6]. The model has one free parameter β fixed by requiring $w_0 \approx -1$. Unlike $f(R)$ gravity, Zx does not modify the Einstein-Hilbert action but adds a geometric term to the energy content.

7 Conclusion

We presented Zx = Z + C + A as a geometric origin of cosmic acceleration [5, 6]. The model fits all current data and makes specific predictions for DESI and Euclid. The 7 figures demonstrate the framework from Planck epoch to present day. Future work will explore Zx in the early universe and structure formation.

A Mathematical Details

A.1 Derivation of Eq. (4)

Starting from the ADM formalism, the lapse function N relates to Z via $N = 1 + \dot{Z}$. Inserting into the Hamiltonian constraint yields the C and A terms [5]. The full derivation spans the Riemann-OS formalism.

A.2 Stability Analysis

The Zx model is stable against perturbations for $\beta < 2$ [6]. Ghosts are absent because Zx enters linearly in the action.

References

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ORCID: https://orcid.org/0009-0003-3319-3822